# Simple AI Chatbot

Ollama + Streamlit

## Project Report

## Project Title

Simple AI Chatbot — a local AI question-answering application built with Ollama and Streamlit.

## What is Ollama?

Ollama is a tool that lets you download and run large language models (LLMs) directly on your own computer, instead of relying on a cloud API. It handles loading the model into memory and exposes a simple local interface (both a command line and an API) that other programs can send prompts to and receive AI-generated responses from.

## What is Streamlit?

Streamlit is a Python framework that turns a plain Python script into an interactive web application with just a few lines of code. It provides ready-made UI components — text inputs, buttons, chat elements so a developer can build a usable interface without writing any HTML, CSS, or JavaScript.

## Which Ollama Model Was Used?

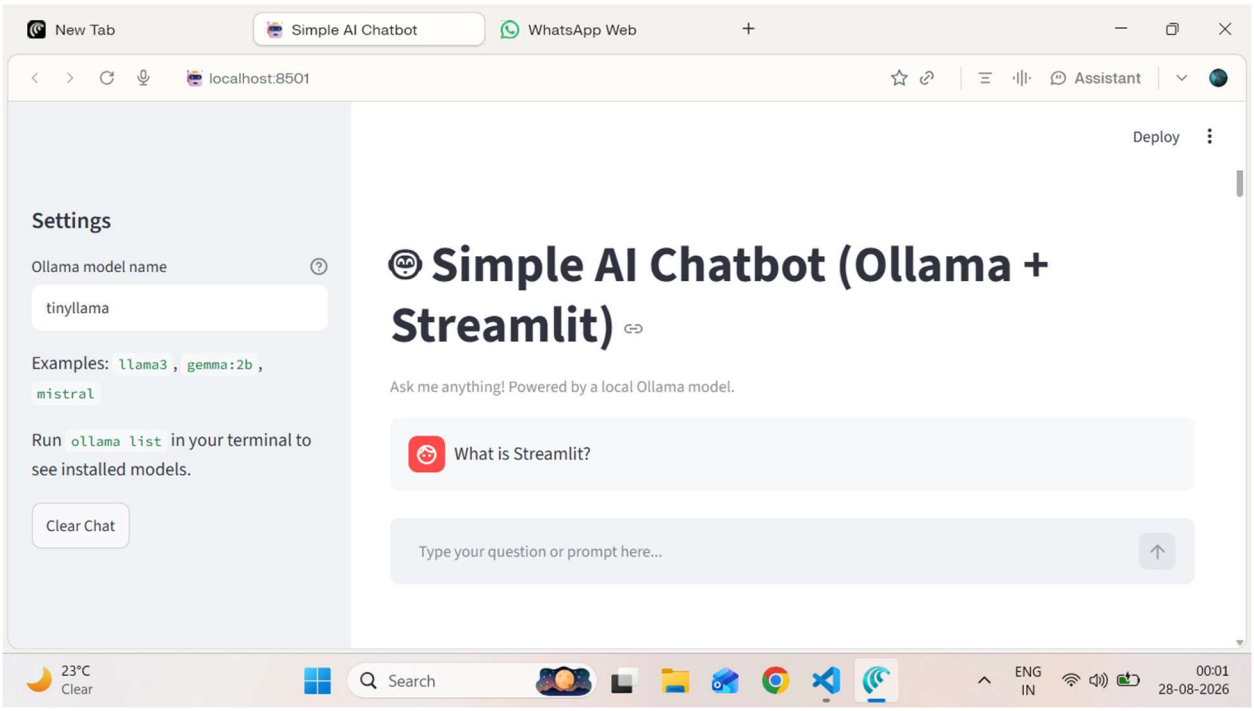
The project used the "tinyllama" model, a lightweight ~1.1B parameter language model. This model was chosen because it runs comfortably on machines with limited RAM (4GB or less) while still being able to hold a basic conversation and answer general questions.

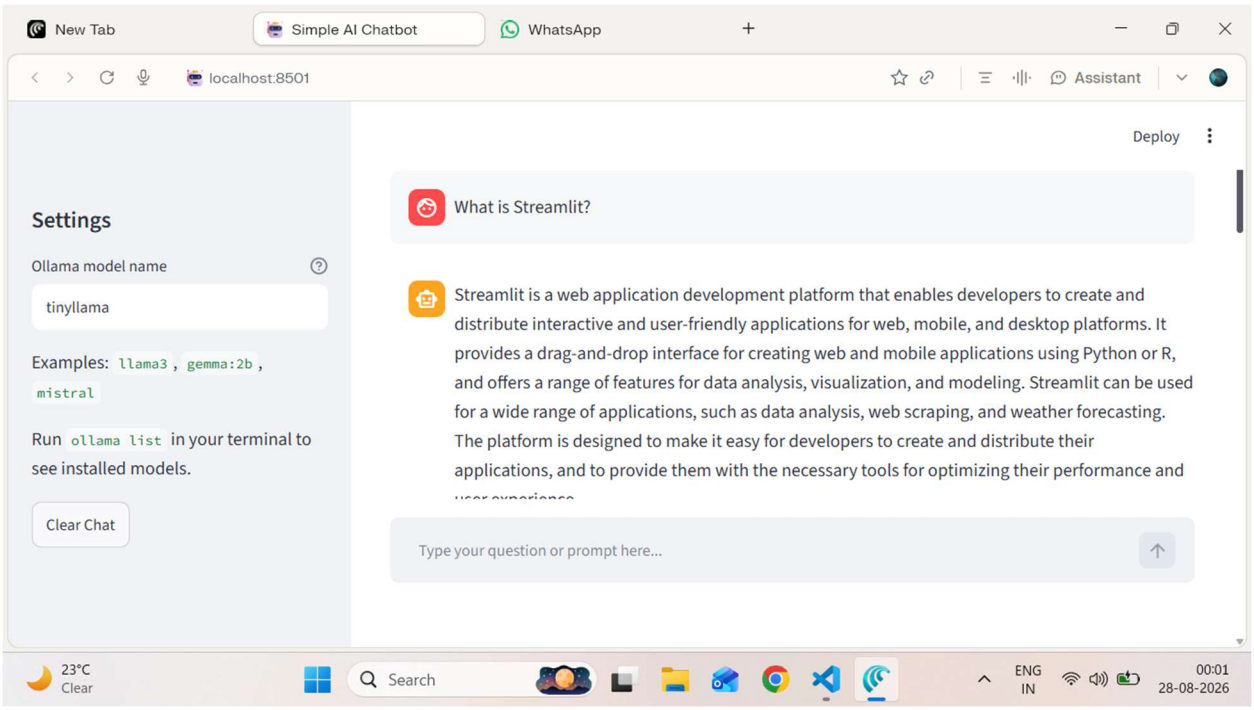
## How the Application Works.

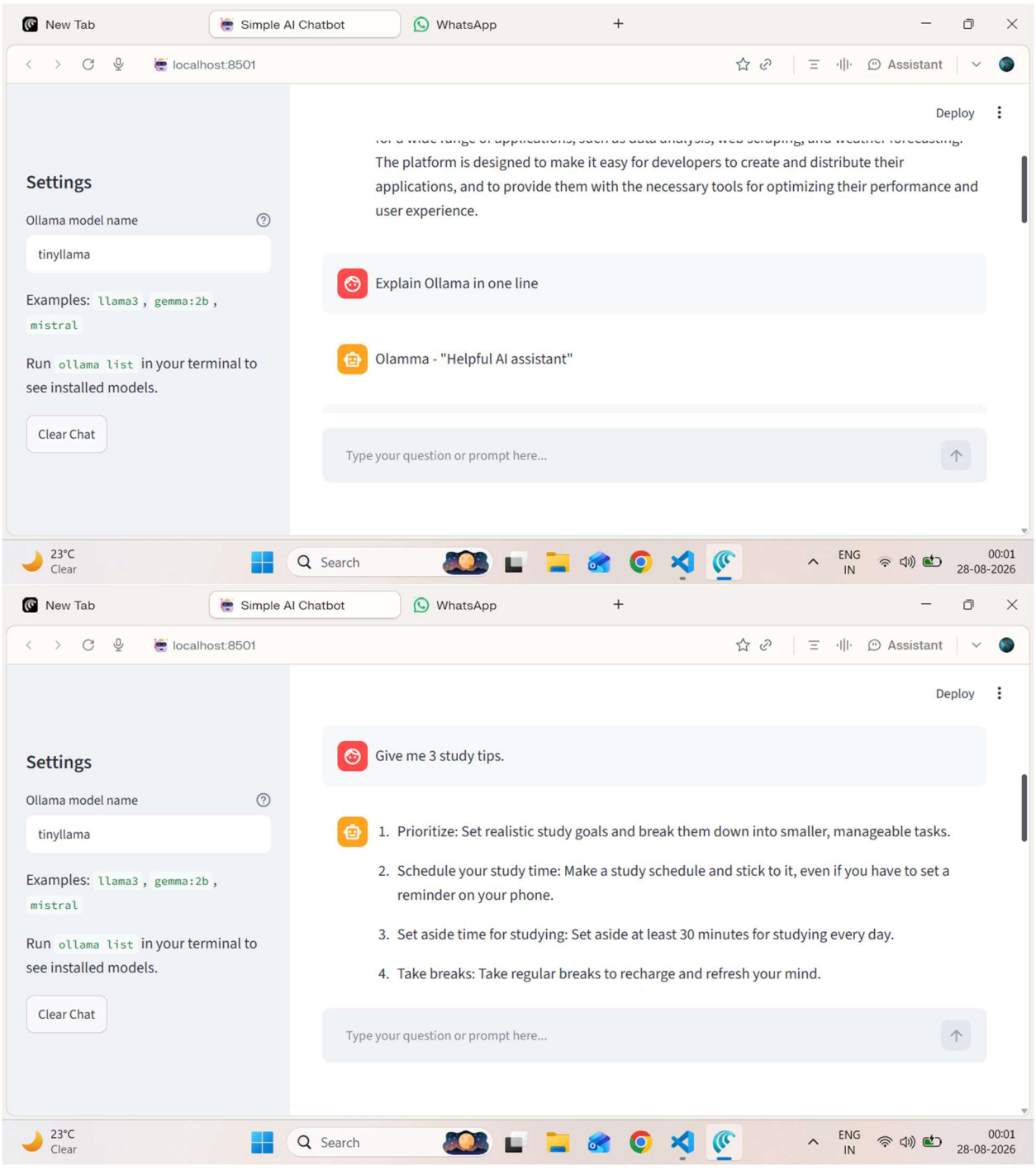
The application (app.py) uses Streamlit to display a simple chat interface with a text input box at the bottom. When the user types a question and presses Enter:

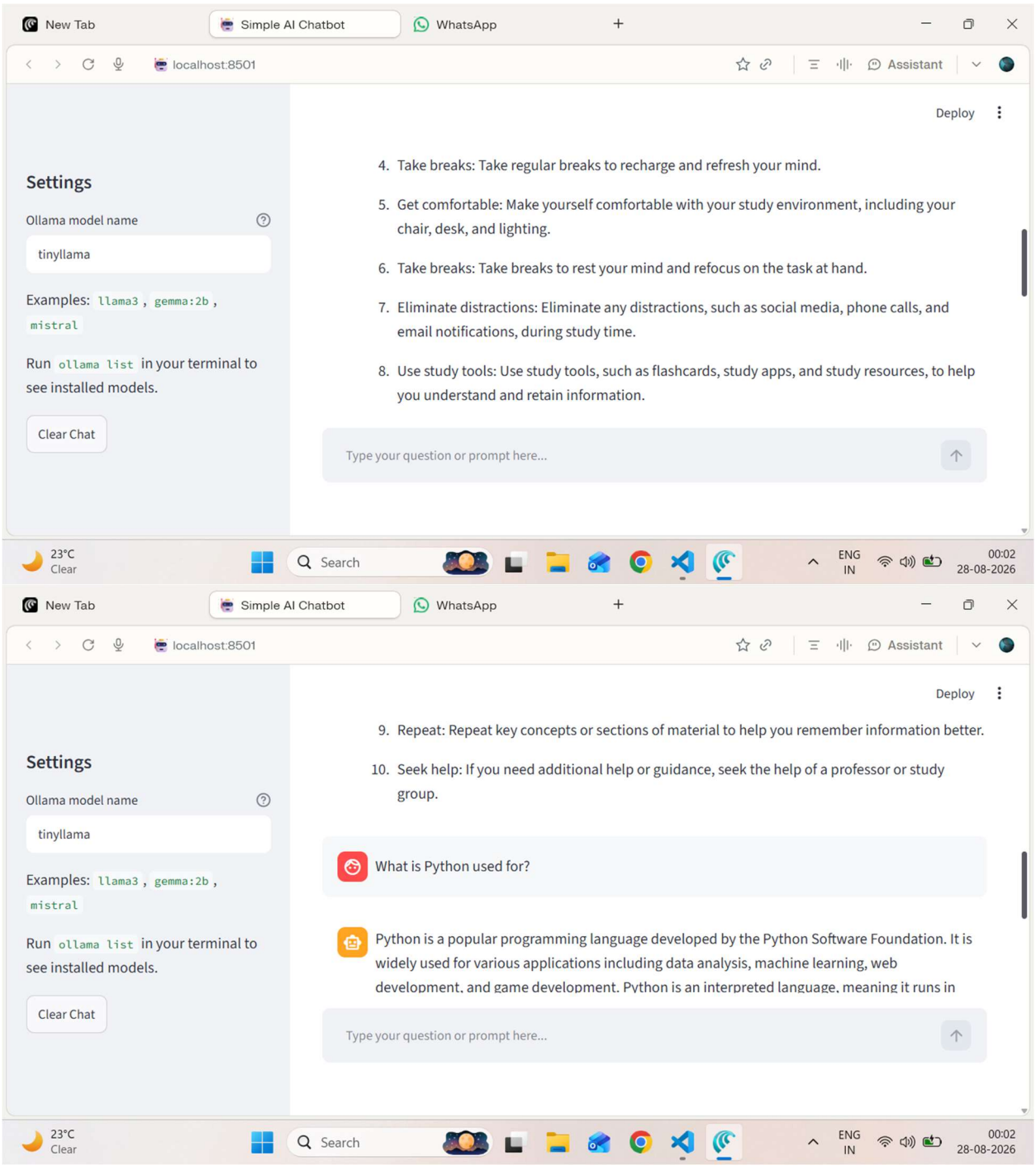
- The prompt is captured by Streamlit's chat_input component.
- It is sent to the local Ollama model (tinyllama) using the Python 'ollama' library.
- Ollama runs the model on the local machine and generates a text response.
- The response is displayed back in the chat window, and the full conversation history is kept on screen using Streamlit's session state.

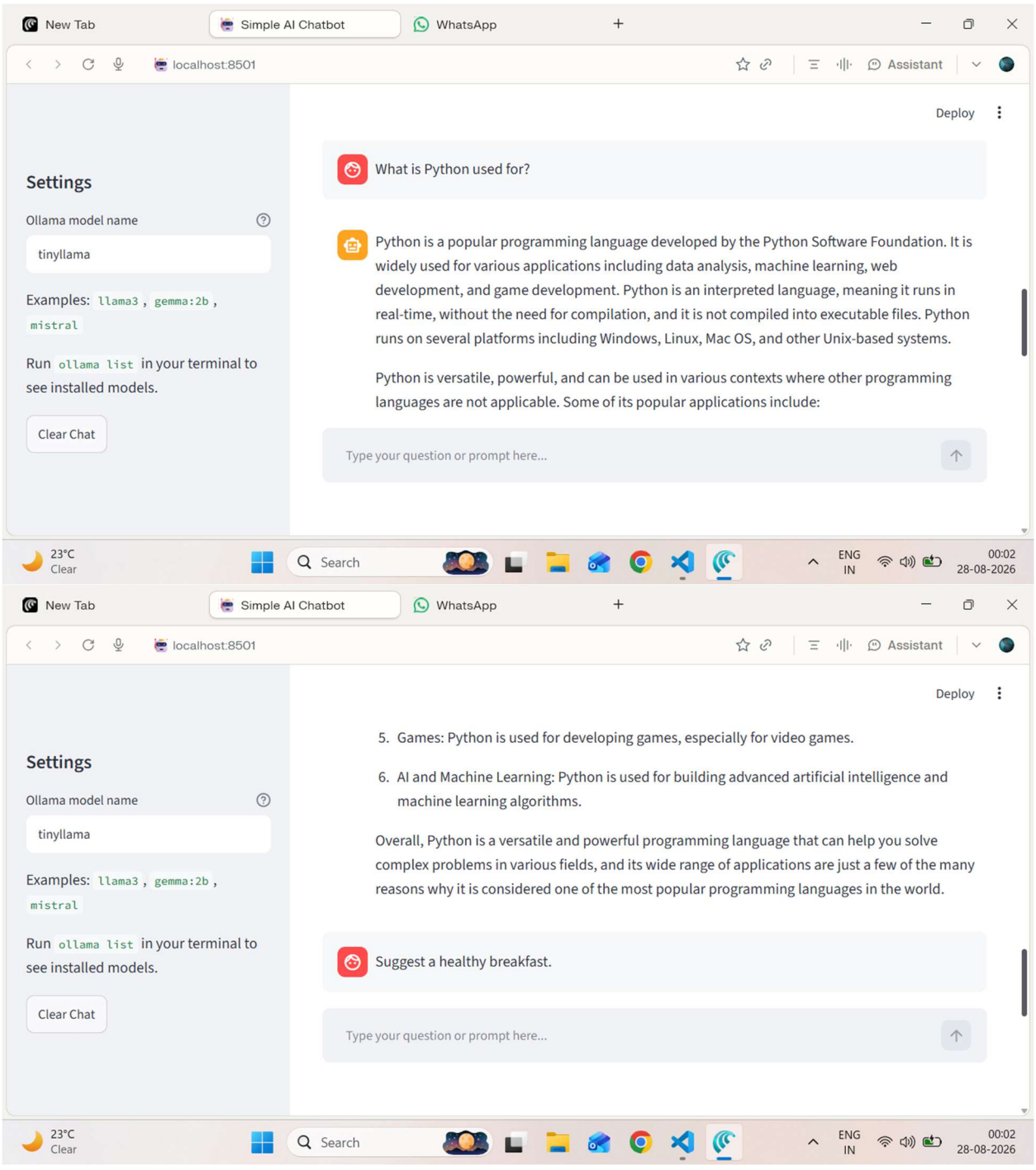
A sidebar lets the user type in a different model name (e.g. switching between tinyllama, gemma:2b, or llama3) without changing any code, and includes a 'Clear Chat' button to reset the conversation.
Screenshots.

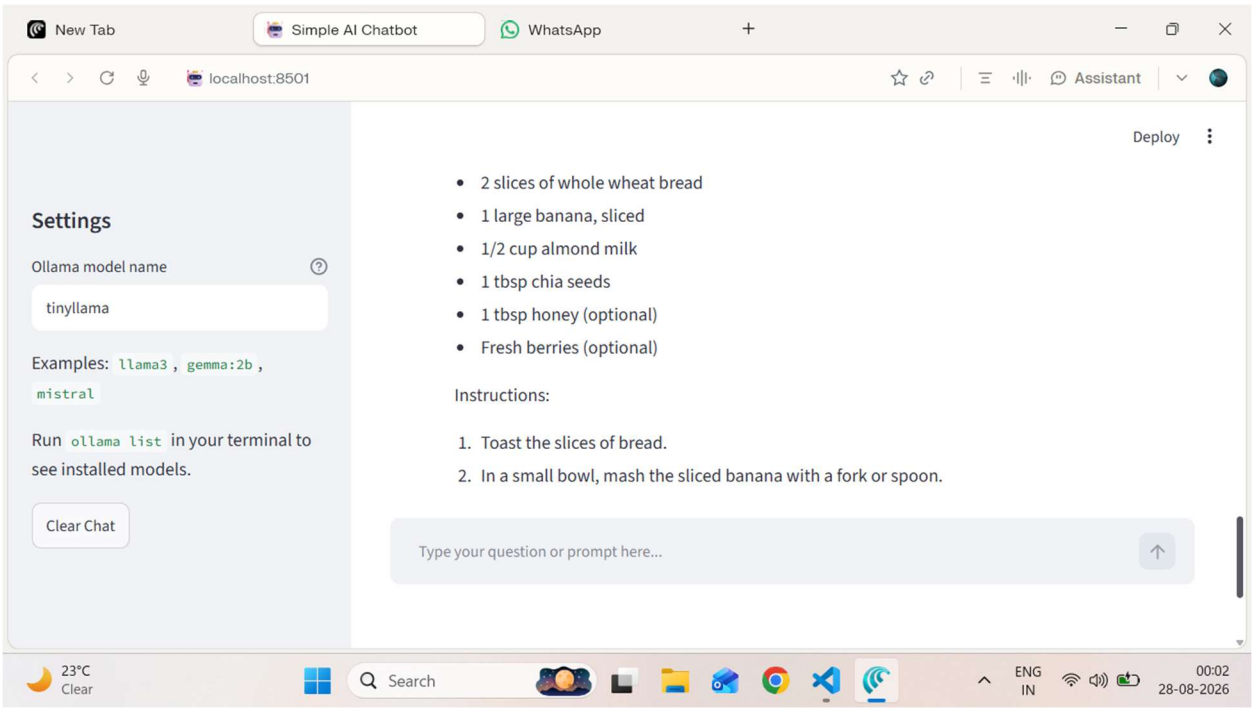

New Tab
Simple AI Chatbot
WhatsApp Web
localhost:8501
Assistant
Deploy
Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
Simple AI Chatbot (Ollama + Streamlit)
Ask me anything! Powered by a local Ollama model.
What is Streamlit?
Type your question or prompt here...
23°C
Clear
Search
ENG
IN
00:01
28-08-2026


New Tab
Simple AI Chatbot
WhatsApp
localhost:8501
Assistant
Deploy
Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
What is Streamlit?
Streamlit is a web application development platform that enables developers to create and distribute interactive and user-friendly applications for web, mobile, and desktop platforms. It provides a drag-and-drop interface for creating web and mobile applications using Python or R, and offers a range of features for data analysis, visualization, and modeling. Streamlit can be used for a wide range of applications, such as data analysis, web scraping, and weather forecasting. The platform is designed to make it easy for developers to create and distribute their applications, and to provide them with the necessary tools for optimizing their performance and
Type your question or prompt here...
23°C
Clear
Search
ENG
IN
00:01
28-08-2026


New Tab
Simple AI Chatbot
WhatsApp
localhost:8501
Assistant
Deploy
Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
The platform is designed to make it easy for developers to create and distribute their applications, and to provide them with the necessary tools for optimizing their performance and user experience.
Explain Ollama in one line
Olamma - "Helpful AI assistant"
Type your question or prompt here...
23°C
Clear
Search
ENG
IN
00:01
28-08-2026
New Tab
Simple AI Chatbot
WhatsApp
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Ollama model name
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Examples: llama3, gemma:2b, mistral
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Clear Chat
Give me 3 study tips.
1. Prioritize: Set realistic study goals and break them down into smaller, manageable tasks.
2. Schedule your study time: Make a study schedule and stick to it, even if you have to set a reminder on your phone.
3. Set aside time for studying: Set aside at least 30 minutes for studying every day.
4. Take breaks: Take regular breaks to recharge and refresh your mind.
Type your question or prompt here...
23°C
Clear
Search
ENG
IN
00:01
28-08-2026


New Tab
Simple AI Chatbot
WhatsApp
localhost:8501
Assistant
Deploy
Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
4. Take breaks: Take regular breaks to recharge and refresh your mind.
5. Get comfortable: Make yourself comfortable with your study environment, including your chair, desk, and lighting.
6. Take breaks: Take breaks to rest your mind and refocus on the task at hand.
7. Eliminate distractions: Eliminate any distractions, such as social media, phone calls, and email notifications, during study time.
8. Use study tools: Use study tools, such as flashcards, study apps, and study resources, to help you understand and retain information.
Type your question or prompt here...
23°C
Clear
Search
ENG
IN
00:02
28-08-2026
New Tab
Simple AI Chatbot
WhatsApp
localhost:8501
Assistant
Deploy
Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
9. Repeat: Repeat key concepts or sections of material to help you remember information better.
10. Seek help: If you need additional help or guidance, seek the help of a professor or study group.
What is Python used for?
Python is a popular programming language developed by the Python Software Foundation. It is widely used for various applications including data analysis, machine learning, web development, and game development. Python is an interpreted language, meaning it runs in
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23°C
Clear
Search
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00:02
28-08-2026


New Tab
Simple AI Chatbot
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localhost:8501
Assistant
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Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
What is Python used for?
Python is a popular programming language developed by the Python Software Foundation. It is widely used for various applications including data analysis, machine learning, web development, and game development. Python is an interpreted language, meaning it runs in real-time, without the need for compilation, and it is not compiled into executable files. Python runs on several platforms including Windows, Linux, Mac OS, and other Unix-based systems.
Python is versatile, powerful, and can be used in various contexts where other programming languages are not applicable. Some of its popular applications include:
Type your question or prompt here...
23°C
Clear
Search
ENG
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00:02
28-08-2026
New Tab
Simple AI Chatbot
WhatsApp
localhost:8501
Assistant
Deploy
5. Games: Python is used for developing games, especially for video games.
6. AI and Machine Learning: Python is used for building advanced artificial intelligence and machine learning algorithms.
Overall, Python is a versatile and powerful programming language that can help you solve complex problems in various fields, and its wide range of applications are just a few of the many reasons why it is considered one of the most popular programming languages in the world.
Settings
Ollama model name
tinyllama
Examples: llama3, gemma:2b, mistral
Run ollama list in your terminal to see installed models.
Clear Chat
Suggest a healthy breakfast.
Type your question or prompt here...
23°C
Clear
Search
ENG
IN
00:02
28-08-2026




## Five Prompts Tested

1. What is Streamlit?
2. Explain Ollama in one line.
3. Give me 3 study tips.
4. What is Python used for?
5. Suggest a healthy breakfast.

## What I Learned from This Project

- How to run a large language model locally using Ollama, without needing a cloud API or internet connection during inference.
- How to build a simple, interactive web UI in Python using Streamlit, including chat components and session state.
- How hardware constraints (like available RAM) directly affect which AI models can realistically run on a given machine, and the trade-off between model size and response quality.
- How to connect a Python backend (Ollama's API) to a frontend (Streamlit) to create a complete, working AI application end-to-end.